

Assignment – Implement Authentication & Authorization

Objective

Enhance the existing **StudentRegistrationWebApp** by implementing **Authentication**, **Authorization**, and **Role-Based Navigation** using **ASP.NET Core Identity**.

Requirements

Part 1 – Authentication

1. Enable **Individual Accounts Authentication** for the application (if not already configured).
2. Create an **Administrator** account manually.
3. Allow new users to **Register** and **Login** using ASP.NET Core Identity.
4. After successful registration, every new user should automatically be assigned the **Student** role.
5. After successful login, display the logged-in user's email in the navigation bar.

Part 2 – Authorization

Implement role-based authorization with the following requirements:

Administrator

The Administrator should be able to:

- View the list of all Courses.
- Add a new Course.
- Edit Course details.
- Delete a Course.
- View the list of all registered Students.

Student

A Student should be able to:

- View the list of available Courses.
- Register for **only one** Course.
- View his/her own profile.
- Edit personal details.
- Logout from the application.

A Student **must not** be able to:

- Create, Edit or Delete Courses.
- View details of other students.
- View the complete Student List.

Part 3 – Navigation Menu

Display navigation links based on the logged-in user's role.

Anonymous User

- Home
- Courses
- Register
- Login

Student

- Home
- Courses
- Register for Course (*display only if the student has not already registered for a course*)
- My Profile
- Logout

Administrator

- Home
- Courses
- Students
- Logout

Part 4 – Security

Apply appropriate authorization attributes to secure the application.

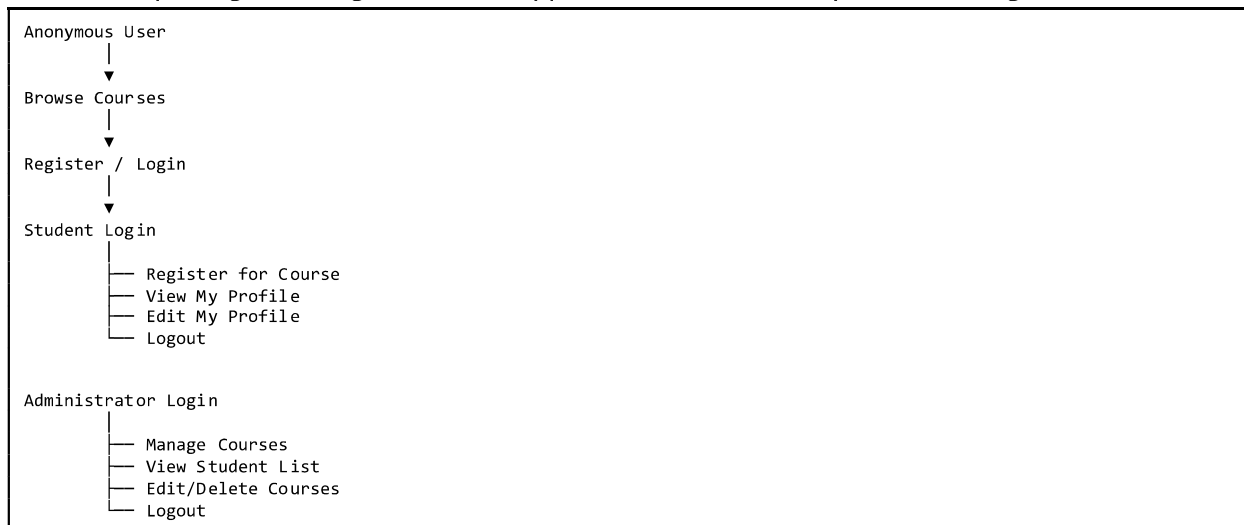
- Anonymous users should be able to view the Home page and Course List.
- Only authenticated Students should be able to register for a course.
- Only Administrators should be able to manage Courses.
- Users should not be able to access restricted pages by typing the URL directly.

Part 5 – User Interface

- Display a welcome message after successful login.
- Display the logged-in user's email on the navigation bar.
- Show appropriate messages for successful registration, login, logout, and course registration.
- Display an **Access Denied** page whenever a user attempts to access an unauthorized page.

Expected Functionality

After completing the assignment, the application should satisfy the following workflow:



Submission Requirements

Students should submit:

- Source Code of the project (**separate .cs file** or in a **single pdf or .docx**).
- Screen shot of SQL Server database containing Identity and application tables.
- A screenshot of UI:
 - Administrator Navigation
 - Student Navigation
 - Anonymous Navigation
 - Successful Login
 - Successful Course Registration
 - Access Denied page
 - Student Profile page

This assignment naturally extends the previous database and MVC assignment into a complete role-based web application while reinforcing Identity, Authentication, Authorization, and role-based UI customization.